

Dynamic Platoon Formation: Spatio-Temporal Matching for Towing-Based Cooperative Autonomous Mobility

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Abstract

We introduce *dynamic platoon formation*, a spatio-temporal matching problem in which larger active vehicles mechanically tow smaller passive vehicles along shared road segments, reducing the self-propelled distance that passive vehicles would otherwise travel independently. Given fixed vehicle trajectories, entry times, speeds, and towing capacities, the problem requires deciding which passive vehicles attach to which active vehicles and over which segments, subject to spatial overlap, point-wise capacity, and temporal synchronisation at coupling points. Unlike standard platooning or one-to-one assignment, the problem allows multi-segment service, where a passive vehicle may be towed by different active vehicles over disjoint portions of its route while active-vehicle capacity is respected at every position along the corridor. We propose two approximation algorithms: a greedy maximum-weight matching baseline and an Iterative Linear Assignment (ILA) method that expands active vehicles into virtual capacity slots and repeatedly solves linear sum assignment subproblems. In synthetic corridor scenarios with up to 400 passive and 80 active vehicles, both methods reduced passive vehicles' self-propelled distance by 25–52% compared with a no-platooning baseline, where every vehicle travels independently. Under a per-distance propulsion model, this reduction serves as a direct proxy for passive-vehicle energy savings. Across all tested scenarios, ILA consistently matched or outperformed the greedy approach in solution quality. At the same time, ILA reduced empirical runtime by a factor of 2.8–11.5. These results suggest that iterative, globally coordinated assignment is both more effective and more scalable than myopic matching for dynamic platoon formation at fleet scale.

CCS Concepts

• **Theory of computation** → **Network optimization**; • **Information systems** → *Spatial-temporal systems*; • **Applied computing** → **Transportation**.

Keywords

Autonomous vehicles, vehicle platooning, spatial-temporal matching, linear assignment, sustainable mobility

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1 Introduction

Single-occupancy travel remains a major barrier to sustainable urban mobility. U.S. National Household Travel Survey data show that drive-alone commuting dominates trip demand [1], while the transportation sector contributes about 29% of U.S. greenhouse gas emissions [2]. This inefficiency is pronounced when many vehicles traverse overlapping routes but still consume propulsion energy independently. With rapid advances in autonomous driving [3], future transportation can realize physically coordinated vehicle platoons that offset a substantial share of the propulsion energy vehicles would otherwise spend driving on their own.

In a near-future setting where vehicles are fully autonomous, new forms of physical coordination between vehicles become feasible: a large autonomous vehicle could mechanically couple with smaller vehicles on a shared road segment, towing them and offsetting a substantial share of their independent propulsion energy [3, 4]. Concrete engineering precedents already exist in patent and public-transport research: a U.S. patent describes autonomous platooning vehicles connected by mechanical linkages controllable by an onboard unit [8], and the Dynamic Autonomous Road Transit (DART) study investigated electronic coupling and decoupling of platooned units along travel corridors [9]. Realizing such coordinated interaction at scale requires *systematic decisions* about which vehicles should couple, when, and for how long along their routes. We therefore treat mechanical coupling, safety, and regulatory design as system assumptions outside the scope of this short paper, and focus on the spatio-temporal matching layer that would be required if such coupling were available.

To address this challenge we introduce a new problem that we call *dynamic platoon formation*. We consider two classes of autonomous vehicles: larger *Active Vehicles* (denoted AVs) capable of physically towing smaller vehicles, and smaller *Passive Vehicles* (denoted PVs) that can be attached to AVs during shared road segments. Unlike drafting-based platooning, which yields modest aerodynamic savings while vehicles remain physically separated, our model assumes mechanical coupling: while a PV is attached to an AV we model the PV as not self-propelled over that segment, with the segment's towing load attributed to the AV. Given a set of AVs and PVs with known, fixed routes (AVs do not deviate from their pre-planned paths), entry times, and speeds, the goal is to

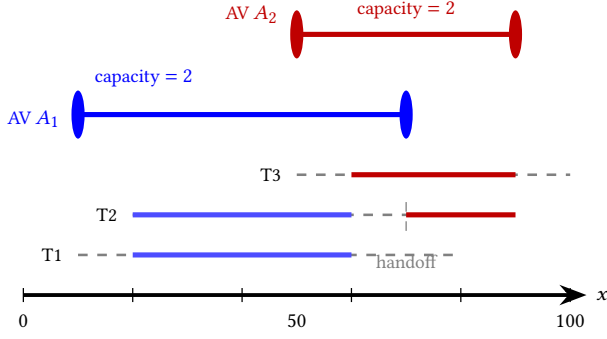


Figure 1: Towing-based cooperation on a 1D corridor (0–100 m). Solid segments are towed travel (PV modeled as not self-propelled); dashed segments are self-driving. AVs A_1, A_2 each have capacity 2; T_2 is handed off from A_1 to A_2 at $x=70$, illustrating *multi-segment matching*.

maximize the total PV distance covered by towing. We use towed distance as a proxy for PV-side propulsion-energy reduction and reserve physics-level load modeling for future work.

Figure 1 illustrates this on a shared corridor with two AVs and three PVs. Two features are central to our formulation: *point-wise capacity* (between $x=20$ and $x=60$, A_1 simultaneously tows T_1 and T_2 , reaching its capacity of 2) and *multi-segment matching*, where T_2 is served by A_1 and A_2 on disjoint portions of its route with a self-driving gap in between.

Prior work falls into three threads: platooning via aerodynamic coordination among physically separated vehicles [4–7], ridesharing and dial-a-ride matching of passengers or trips [10–13], and combinatorial one-to-one assignment [14, 15]. None captures the combination required here: aerodynamic platooning does not model mechanical coupling, ridesharing matches people rather than vehicles and does not enforce point-wise capacity along a corridor, and one-to-one assignment cannot represent a single PV being served by multiple AVs over disjoint segments. Dynamic platoon formation requires all three features simultaneously, making it a distinct combinatorial problem.

We formulate the problem as: given AVs and PVs with fixed routes, entry times and speeds, decide which PVs attach to which AVs over which road segments to maximize the total towed PV distance, subject to spatial overlap requirements, point-wise capacity, and temporal synchronization at coupling points. This yields a state-dependent combinatorial matching problem that differs from a static Generalized Assignment Problem (GAP) [16], because assigning one towing segment can change later temporal feasibility for the same PV. Exact global solvers are therefore impractical at fleet scale, and we propose two approximations. The first, a *Greedy Maximum-Weight Matching*, repeatedly commits to the highest-saving feasible AV–PV segment until no feasible pair remains; this approach is fast but myopic. The second, an *Iterative Linear Assignment* (ILA) algorithm, takes a globally coordinated view: in each round it expands every AV into virtual capacity slots, builds a bipartite graph of feasible AV-slot–PV pairings, and solves the resulting *Linear Sum Assignment Problem* (LSAP) optimally via the

Jonker–Volgenant method [17]. PV states are then updated to reflect newly covered segments and the round repeats until no further assignment is possible. For tractability we evaluate both algorithms on a one-dimensional corridor abstraction; we discuss extensions to 2D road networks in Section 5.

Contributions. (1) We formalize *dynamic platoon formation* as a spatio-temporal matching problem with multi-segment service and point-wise AV capacity, instantiated on a 1D corridor. (2) We develop and compare two approximation algorithms: a greedy maximum-weight baseline and an iterative LSAP-based method with virtual-slot capacity encoding. (3) On synthetic corridor scenarios with up to 80 AVs and 400 PVs, ILA matches or exceeds greedy solution quality (25–52% towed-distance coverage depending on capacity) while running 2.8–11.5× faster.

2 Problem Formulation

We formalize *dynamic platoon formation* on a one-dimensional corridor of length L . Restricting attention to a single corridor isolates the spatio-temporal matching structure (segment overlap, point-wise capacity, and temporal synchronization) from routing decisions; extending the formulation to 2D road networks is discussed in Section 5. Throughout, positions are in meters (m), times in seconds (s), and speeds in m/s; towing capacities are nonnegative integers.

2.1 Vehicles and Notation

A problem instance consists of N Active Vehicles (AVs) $\mathcal{A}$ and M Passive Vehicles (PVs) $\mathcal{P}$ on a corridor of length L (m). Each AV $a_i \in \mathcal{A}$ is described by $(e_i^a, x_i^a, C_i, t_i^a, v_i^a)$, where $e_i^a, x_i^a \in [0, L]$ are entry/exit positions in m with $e_i^a < x_i^a$, $C_i \in \mathbb{Z}_{\geq 0}$ is a scalar towing capacity, t_i^a is the entry time in s, and $v_i^a > 0$ is a constant speed in m/s. Each PV $p_j \in \mathcal{P}$ is described analogously by $(e_j^p, x_j^p, t_j^p, v_j^p)$ but has no towing capacity. Two instance-level parameters complete the input: a minimum towing-segment length $L_{\min}$ (m) and a temporal coupling tolerance τ (s). The capacity *ranges* reported in Section 4 are values swept across instances, not per-route vectors for a single AV.

2.2 Feasibility of a Towed Segment

A candidate towing segment is a tuple (a_i, p_j, s, e) with $0 \leq s < e \leq L$, meaning p_j is towed by a_i over $[s, e]$. The maximum realizable overlap of (a_i, p_j) is $[cp_{ij}, dp_{ij}]$ with $cp_{ij} = \max(e_i^a, e_j^p)$ (earliest possible coupling) and $dp_{ij} = \min(x_i^a, x_j^p)$ (latest possible decoupling); its length is $d_{ij} = \max(0, dp_{ij} - cp_{ij})$, and any selected segment must satisfy $cp_{ij} \leq s < e \leq dp_{ij}$. The segment is *feasible* if (i) $[s, e] \subseteq [cp_{ij}, dp_{ij}]$, (ii) $e - s \geq L_{\min}$ (minimum length, justifying coupling overhead), and (iii) the two vehicles reach s within tolerance τ :

$$|T_i^a(s) - T_j^p(s)| \leq \tau, \quad (1)$$

where $T_i^a(x) = t_i^a + (x - e_i^a)/v_i^a$ is AV a_i 's arrival time at position x . The PV arrival time $T_j^p(s)$ is *state-dependent*: for an initially uncovered PV it is the self-driving time $t_j^p + (s - e_j^p)/v_j^p$, but after p_j has been towed on an earlier segment, our algorithms (Section 3) update its route-state time before evaluating later segments. We use the towed distance $e - s$ directly as the saving proxy on a feasible

segment, reserving energy-level modeling that accounts for AV-side towing load for future work.

2.3 Multi-Segment Matching and Optimization

Let $\mathcal{X}$ denote a set of feasible segments selected by the algorithm. We allow a single PV to be served by multiple AVs along disjoint portions of its route (*multi-segment matching*), but the same PV cannot be towed by two AVs at the same position. The optimization problem maximizes total towed distance

$$\max_{\mathcal{X}} \sum_{(a_i, p_j, s, e) \in \mathcal{X}} (e - s) \quad (2)$$

subject to three conditions: (i) the *point-wise capacity* constraint

$$|\{(a_i, p_j, s, e) \in \mathcal{X} : x \in [s, e)\}| \leq C_i \quad \forall a_i \in \mathcal{A}, \quad \forall x \in [e_i^a, x_i^a], \quad (3)$$

which bounds, at every position x along AV a_i 's route, the number of PVs currently attached to a_i by C_i ; (ii) *PV non-overlap*, meaning that segments assigned to the same PV have disjoint spatial intervals, so a PV is never towed by two AVs at the same position; and (iii) *per-segment feasibility*, meaning that every selected segment satisfies the overlap, minimum-length, and temporal-synchronization conditions in Section 2.2

Problem structure. The problem has a combinatorial assignment structure with *state-dependent* feasibility: selecting one towing segment changes a PV's residual route timing and can affect the temporal feasibility of later segments, distinguishing it from a static GAP-style assignment [16]. We therefore do not pursue an exact global solver in this short paper and instead study scalable approximation algorithms in Section 3.

Critical-points reduction. Although constraint (3) ranges over a continuum of positions x , the count of attached PVs is a step function that changes only at the endpoints of selected segments. It therefore suffices to enforce capacity at the finite set of *critical points* $C_i = \{e_i^a, x_i^a\} \cup \{s, e : (a_i, p_j, s, e) \in \mathcal{X}\}$ for each AV a_i . Both algorithms in Section 3 exploit this reduction so that each feasibility check costs $O(|C_i|)$ rather than continuous verification.

2.4 Modeling Assumptions

We adopt the following modeling choices, each open to relaxation in future work. Routes, entry times, and speeds are known and deterministic, and AVs do not re-route to attract PVs; while attached, a PV follows the AV's speed. Coupling and decoupling are modeled as instantaneous and cost-free at the algorithmic layer, with the physical overhead partly absorbed by the minimum-segment requirement $L_{\min}$. Because matching does not change the total vehicle population, we do not model induced congestion in this short paper, and mechanical, safety, and regulatory aspects of physical coupling remain system-level assumptions outside our scope (cf. Sec. 1).

3 Proposed Algorithms

Both algorithms maintain a PV *route-state* $(\ell_j, \hat{t}_j)$ where ℓ_j is the start of the unprocessed suffix of p_j 's route (initially e_j^p) and $\hat{t}_j$ its arrival time at ℓ_j (initially t_j^p). When p_j is committed to a segment $[s, e]$ on AV a_i , the state advances by $\ell_j \leftarrow e$ and $\hat{t}_j \leftarrow \hat{t}_j + (s - \ell_j^{\text{old}})/v_j^p + (e - s)/v_i^a$, reflecting self-driving to s followed by towing

at the AV's speed to e . The state-dependent arrival time T_j^p in Eq. (1) reads from this $\hat{t}_j$.

3.1 Greedy Maximum-Weight Matching

The greedy baseline repeatedly enumerates feasible segments using the critical-point capacity check (Sec. 2.3), commits the highest-saving segment ($e - s$), updates the route-state, and stops when no feasible segment remains. Sorting $O(NM)$ candidates per iteration gives $O((NM)^2 \log NM)$ total time; the method is fast but myopic, since one high-saving commit can block a co-optimal alternative.

3.2 Iterative Linear Assignment (ILA)

ILA replaces the myopic commit with a sequence of *global* matching rounds. In each round, every AV a_i is replicated into C_i identical *virtual slots*, encoding per-AV multiplicity in the LSAP while leaving the position-dependent capacity check to the commit step. With $K = \sum_i C_i$ slot nodes on the left and M PV nodes on the right, we form a cost matrix $W \in \mathbb{R}^{K \times M}$ whose entry for slot k of AV a_i and PV p_j is $-d_{ij}$ for a feasible pair and $+\infty$ otherwise, where $d_{ij} = \max(0, dp_{ij} - \max(e_i^a, \ell_j))$ is the residual overlap that uses ℓ_j in place of e_j^p . Minimum-cost matching on W therefore maximizes total towed distance for the current round.

Each round (Algorithm 1) solves this Linear Sum Assignment Problem (LSAP) optimally with the Jonker–Volgenant algorithm [14, 17]. Returned pairs are then sequentially confirmed against point-wise capacity at the critical points; pairs that fail (because multiple slots of the same AV matched a non-disjoint set of segments in one round) are discarded and remain eligible in later rounds. Confirmed pairs advance the corresponding PV route-states, so the next round operates on the residual problem. Rounds repeat until no feasible match remains, yielding multi-segment service whenever the same PV is matched to different AVs across rounds. Jonker–Volgenant guarantees per-round optimality on the constructed bipartite subproblem; global optimality across rounds is not guaranteed, but our experiments (Section 4) show ILA matches or exceeds greedy on every tested scenario. Each round runs in $O(KM + \max(K, M)^3)$, and at most $O(NM)$ rounds are required since each commit strictly advances some ℓ_j , giving overall complexity $O(NM \cdot \max(K, M)^3)$. **Algorithm 1 (ILA).**

Require: AVs $\mathcal{A}$, PVs $\mathcal{P}$, $L_{\min}$, τ

Ensure: $\mathcal{X}$

```

1:  $(\ell_j, \hat{t}_j) \leftarrow (e_j^p, t_j^p) \quad \forall j; \quad \mathcal{X} \leftarrow \emptyset$ 
2: repeat
3:   Build  $W$  on  $K = \sum_i C_i$  virtual slots;  $\mathcal{M} \leftarrow \text{LSAP}(W)$ 
4:   for  $(\text{slot}: a_i, p_j) \in \mathcal{M}$  with  $W_{\text{slot}, j} < +\infty$ , in order do
5:      $s \leftarrow \max(e_i^a, \ell_j)$ ,  $e \leftarrow dp_{ij}$ 
6:     if  $(a_i, p_j, s, e)$  feasible w.r.t. prior commits this round
7:       then
8:          $\mathcal{X} \leftarrow \mathcal{X} \cup \{(a_i, p_j, s, e)\}$ ; update  $\ell_j, \hat{t}_j$ 
9:       end if
10:    end for
11: until no commit in the last round
12: return  $\mathcal{X}$ 
```

4 Experimental Evaluation

4.1 Setup

Both algorithms are implemented in Python 3.11 with NumPy 1.26 and SciPy 1.11 [17]; ILA's per-round LSAP uses SciPy's compiled Jonker–Volgenant routine. Experiments run single-threaded on an Apple M4 Pro (16 GB unified memory, macOS 15). We reuse L for corridor length, N and M for AV/PV counts, C_i for AV capacity, $L_{\min}$ for minimum towing length, and τ for temporal tolerance; W denotes the scenario time window for sampling entry times. For synthetic corridor instances, AV and PV entry/exit positions are sampled uniformly from $[0, L]$, entry times uniformly from $[0, W]$, and speeds uniformly within $\pm 20\%$ of a unit reference. Default parameters are $L=100$ m, $N=50$, $M=200$, $L_{\min}=10$ m, $\tau=5$ s, $W=100$ s, with C_i drawn uniformly from $\{1, 2, 3\}$. We report means over four seeds (42–45) and run two sweeps. The *capacity sweep* tests $C_{\max} = 2, 4, 8, 16$ at default N, M, L , sampling C_i uniformly from $\{1, \dots, C_{\max}\}$. The *length sweep* tests six lengths from 50 to 1600 m at $N=80$, $M=400$, $C_{\max}=3$, with $W=L$ s so that vehicle density per unit time stays comparable. The headline metric is *saving %*, the total towed distance divided by the no-towing baseline $\sum_j (x_j^p - e_j^p)$; we also report wall-clock runtime.

Figure 2 reports both sweeps. As $C_{\max}$ grows from 2 to 16, saving % rises from 24.5% to 51.4% for Greedy and from 24.8% to 52.3% for ILA, an ILA advantage of 0.2–0.9 percentage points across the four capacity settings; as L grows from 50 m to 1600 m, saving % falls from 31.4% to 7.4% (Greedy) and 31.6% to 7.5% (ILA), reflecting the diminishing pairwise overlap on longer corridors. ILA matches or exceeds Greedy in every tested configuration. The two algorithms are close in solution quality on a 1D corridor because the linear route structure restricts the combinatorial diversity of feasible matchings; in 2D road networks with richer overlap patterns we expect Greedy's myopic commits to block more globally superior assignments, widening ILA's margin. Wall-clock runtime in our implementation is 2.8–11.5 $\times$ lower for ILA on the capacity sweep and 5.5–26.9 $\times$ lower on the length sweep, attributable to SciPy's compiled LSAP solver and to Greedy's repeated $O(NM \log NM)$ candidate sorting; we report this as an empirical implementation observation rather than a theoretical claim, since worst-case complexity favors Greedy (Section 3). Two limitations bound our interpretation: (i) towed distance is a proxy for PV propulsion-energy reduction and ignores the AV-side towing load, and (ii) the offline, full-information setting assumes deterministic trajectories. Section 5 discusses the 2D extension and a stochastic online formulation as the natural next steps.

5 Conclusion and Future Work

We formalized *dynamic platoon formation* as a spatio-temporal multi-segment matching problem with point-wise AV capacity constraints and demonstrated that an iterative LSAP-based solver consistently matches or outperforms a greedy baseline across all tested 1D corridor configurations. A natural direction for future work is extending the framework to 2D road networks, where each PV may choose among multiple feasible routes rather than following a fixed corridor. This generalization transforms the problem from segment overlap on a single path to overlap on a graph with shared edges, coupling platoon formation with route-choice decisions. Additional

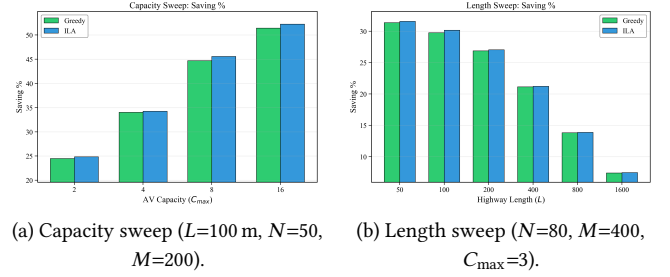


Figure 2: Saving % under (a) varying AV capacity upper bound $C_{\max}$ and (b) varying corridor length L . ILA matches or exceeds Greedy in every tested configuration. Bars are 4-seed means.

research opportunities include a stochastic online setting in which vehicle arrivals and route information are revealed incrementally over time, as well as the incorporation of energy and incentive models that account for AV towing loads and operator participation costs. Such extensions would enable optimization objectives that explicitly balance PV energy savings against AV operational overhead while fostering incentive-compatible participation across multiple operators, rather than relying solely on distance-based proxies.

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